

3 Gentamicin is an antibiotic used to treat severe bacterial infections in children.

- (a) Some children have a genetic mutation in the gene *MT-RNR1*. If gentamicin is given to children with this genetic mutation, it can cause deafness.

Before gentamicin can be given to a child with a severe bacterial infection, PCR (polymerase chain reaction) and electrophoresis are used to test whether the child has this mutation. If the mutation is found, a different antibiotic must be given.

- (i) Describe **and** explain the role of *Taq* polymerase in PCR.

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- (ii) PCR with primers specific to the *MT-RNR1* gene is used to amplify DNA from the child that is being tested.

The PCR primers are designed so that the amplified product of the normal allele of *MT-RNR1* is longer than the amplified product of the mutant allele.

Gel electrophoresis is used to separate the PCR products.

Fig. 3.1 shows the results of gel electrophoresis after using this method of PCR on DNA samples collected from three children.

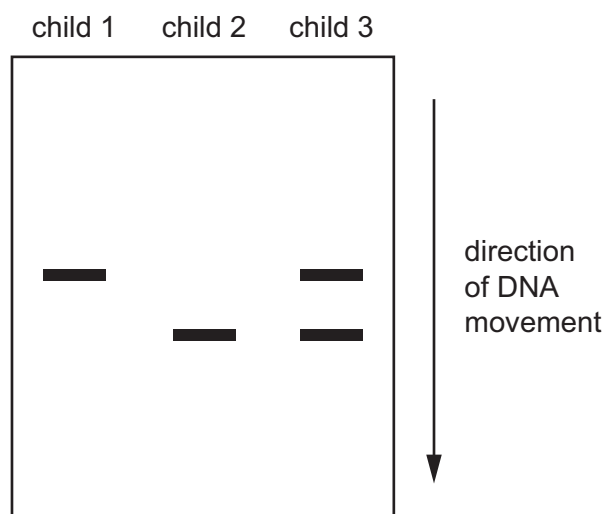


Fig. 3.1



State which of the children in Fig. 3.1 **cannot** receive the antibiotic gentamicin to treat a severe bacterial infection.

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- (iii) Explain how gel electrophoresis produces the pattern of results shown in Fig. 3.1 from the PCR products of the *MT-RNR1* gene.

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- (b) Some bacteria have plasmids that contain a gene conferring resistance to gentamicin. The gene can be transferred to other bacteria.

Suggest how the gentamicin-resistance gene can be transferred to other bacteria.

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[Total: 10]